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FORECAST-CLSTM: A New Convolutional LSTM Network for Cloudage Nowcasting

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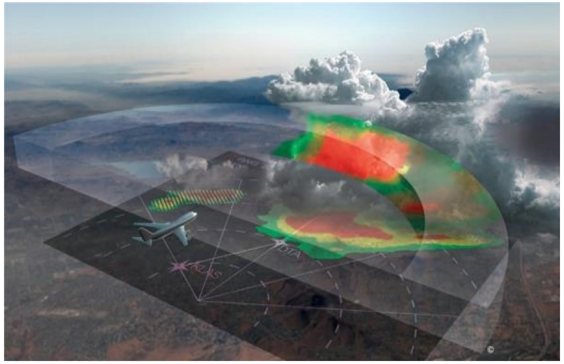
Weather Nowcasting Tasks

Weather Nowcasting: Make detailed forecast of local weather in a short time in the future (generally 0-3 hours), and give early warning of possible abnormal weather phenomena.

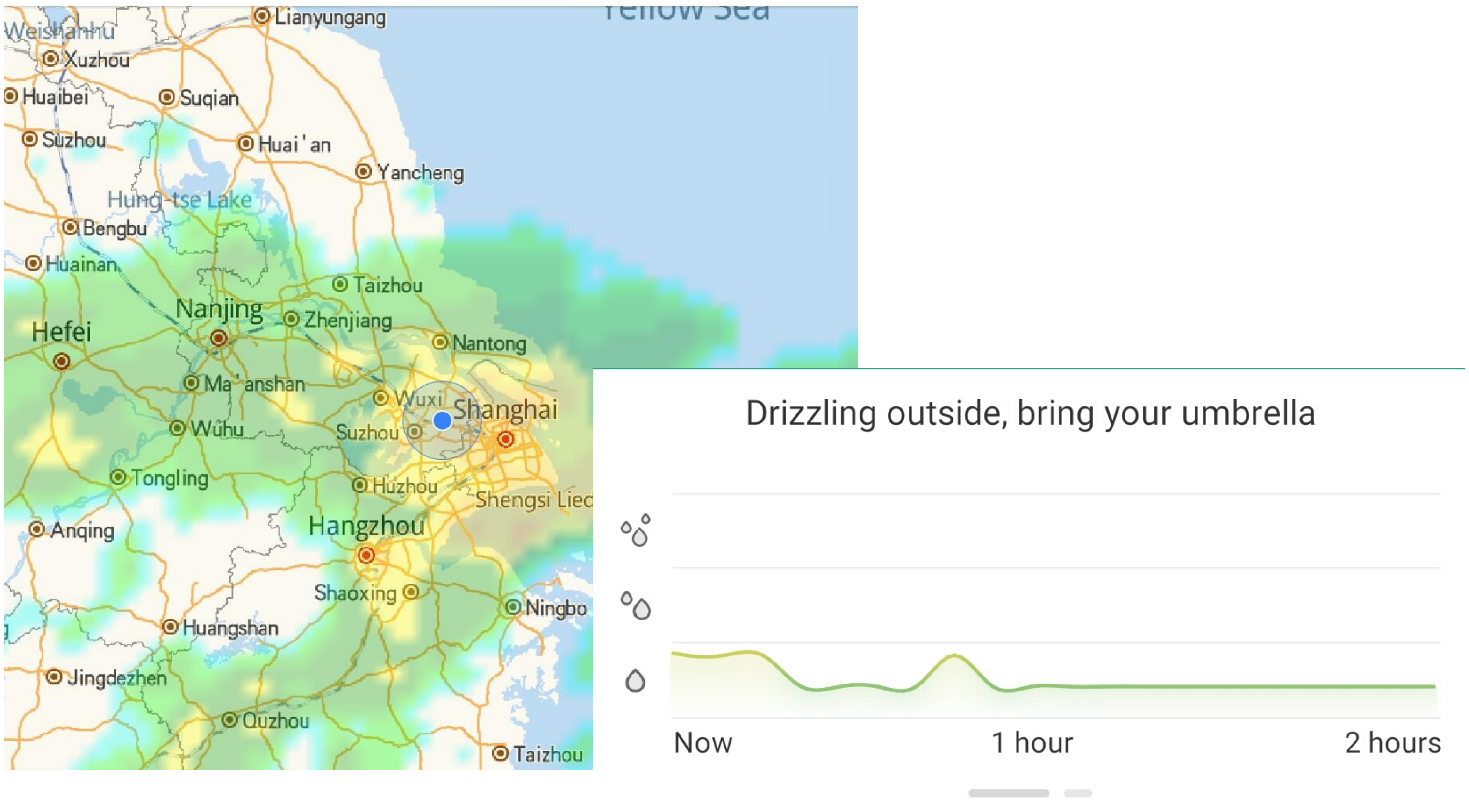
a. outdoor sports weather support

b. urban traffic weather support



c. flight weather support

Nowcasting: High Temporal and Spatial Resolution Prediction



Colorful Weather and China Meteorological Administration(CMA) jointly developed a minute-level high precision radar precipitation nowcasting system.

Applications of Cloudage Nowcasting

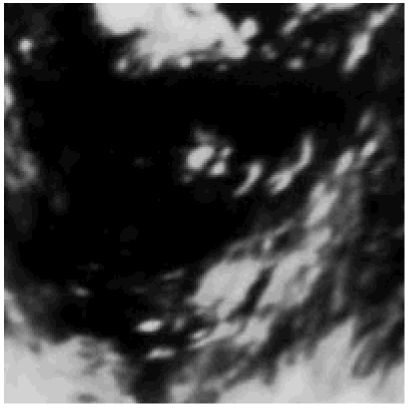
a. agricultural guidance

b. astronomical observations

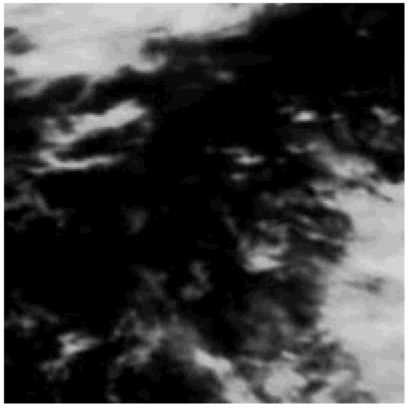
c. ultraviolet index

Challenges: Cloudage Nowcasting Dataset

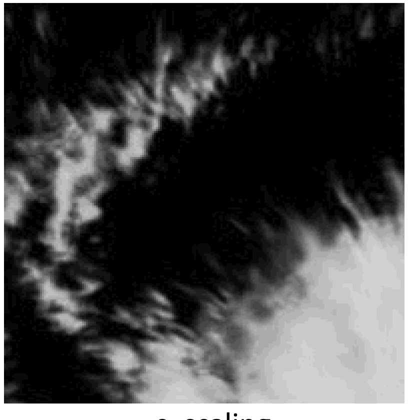
SCMD: The first large-scale **S**atellite **C**loudage **M**ap **D**ataset for deep learning research



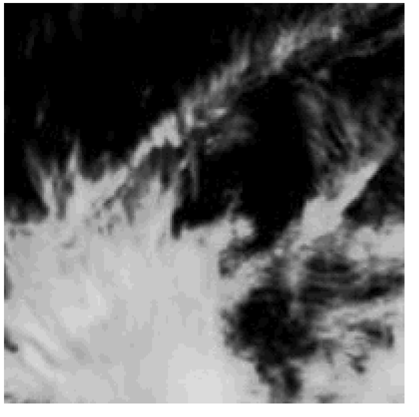
a. rotation



b. translation



c. scaling



d. more complex condition

Challenges: Highly Complex Internal Variations of Weather System

- The variety of internal variations make weather system complex and unpredictable.
- It is hard to predict the internal variation of clouds only by satellite cloud images.
- Further information is needed for helping cloudage nowcasting, such as pressure and temperature information.

Cloudage Nowcasting and Natural Scene Prediction

Difference:

- The features of cloudage images are more complex and diverse, and difficult to abstract at a higher level.
- In terms of the actual weather prediction, cloudage nowcasting task is more concerned with the accuracy of pixel values.

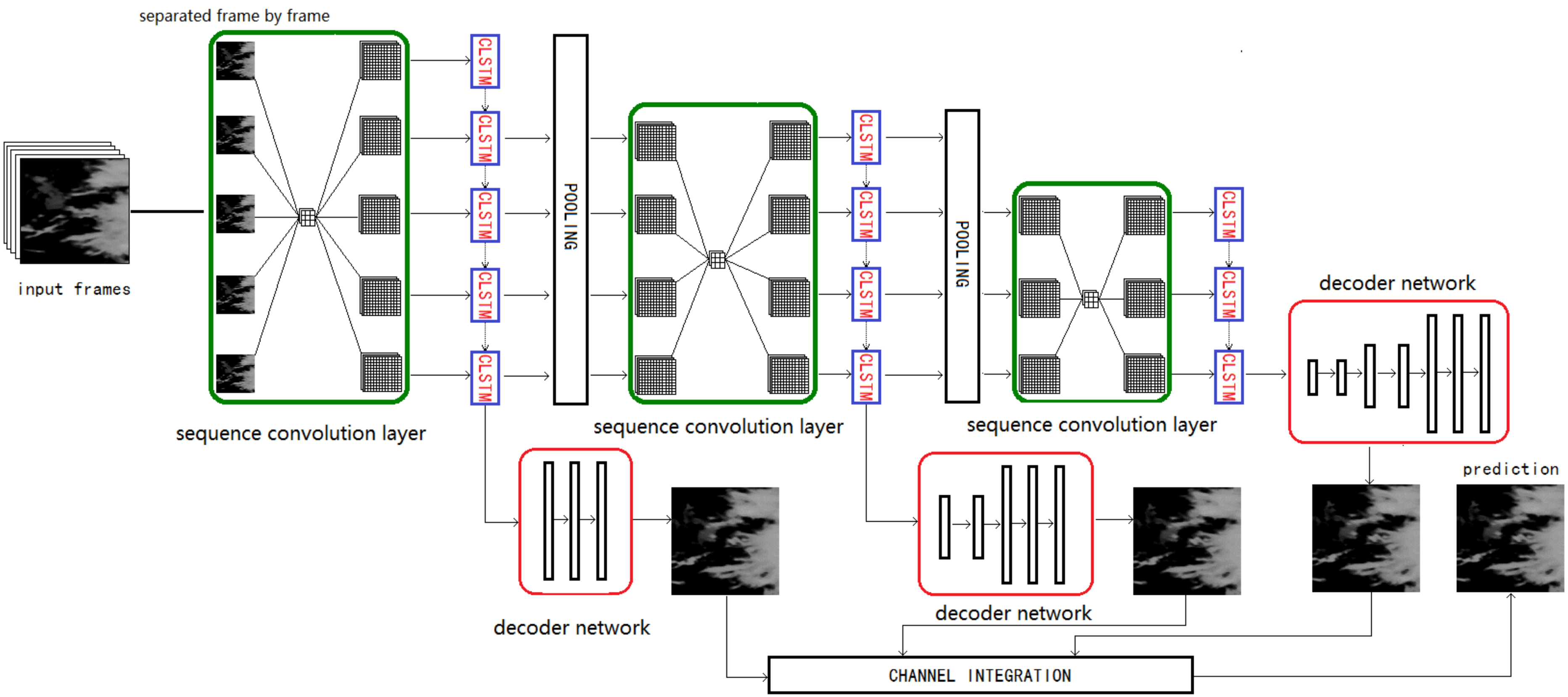
Motivation

- **Deep Learning Based Model**
  Cloudage nowcasting task can be reduced to the of spatio-temporal prediction problem, so it can be handled by deep learning based approach.

- **Convolutional Long-Short-Term Memory Network**
  Convolutional Long-Short-Term Memory network take both spatial and temporal variations into consideration.

- **Hierarchical Predictions**
  Considering the timing changes of features at different levels to improve model performance and reduce information loss.

Network Structure



Architecture of the FORECAST-CLSTM Model. The green box represents the sequence convolution layer and the red box represents the decoding network.

Objective Function

Forecaster Objective Function:

$$loss^t = \sum_{i=1}^{M} \sum_{j=1}^{N} e^{1-\frac{\min(|y_{ij}^t|,|\hat{y}_{ij}^t|)}{255}} \times \left(y_{ij}^t - \hat{y}_{ij}^t\right)^2$$

- mean square error based.
- assigns higher weights to pixels with lower value.
- more suitable for actual weather prediction applications.

Results Comparison

TABLE I
PERFORMANCE OF DIFFERENT MODELS

Model	MSE	PSNR	SSIM	ECCR
Experiments on Moving Mnist++ Dataset				
CLSTM	221.7520	22.3217	0.8106	-
F-CLSTM	70.9074	31.6401	0.9620	-
Experiments on SCMD16 Dataset				
MLP	377.2251	24.0012	0.9305	-
FC-LSTM	359.4351	24.5127	0.9377	-
CLSTM	143.1854	27.6126	0.9744	-
F-CLSTM	113.2776	29.6724	0.9800	54.8%
F-CLSTM-F	114.2779	29.6578	0.9798	55.9%

Effective **C**loud **C**overage **R**atio(ECCR): the ratio of the pixel with cloud presence in the image to the entire space.

Conclusion

- Convolutional Long-Short-Term Memory network is used to predict spatiotemporal sequences.
- Making hierarchical predictions to cope with the variability of low-level features and reduce information losses.

- Using the pixel value related objective function to output more realistic weather prediction results.

THANKS